SM Graded Quiz M3

Question 1

How does the quad function contribute to the Lewis pricing implementation?

- It generates random numbers
- It optimizes parameters
- It calculates Greeks
- It numerically integrates the characteristic function

Question 2

What is the significance of the Lewis (2001) approach in this implementation?

- It eliminates the need for calibration
- It enables semi-analytical option pricing using characteristic functions

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- It only works for European puts
- It provides Monte Carlo simulation

Question 3

Why does the code check if ρ is between -1 and 1 in the error function?

- To match market data
- To ensure correlation is mathematically valid
- To speed up calculation
- To prevent arbitrage

Question 4

Why does the error function return 500.0 when the Feller condition $(2\kappa\nu\theta\nu < \sigma\nu^2)$ is violated?

- To speed up computation
- To penalize parameter sets that would make variance negative

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- To ensure faster convergence
- To match market convention

Question 5

What is the key difference between the error metrics used in Heston vs. Merton calibration as shown in the material?

- Merton uses MAE while Heston uses MSE
- Heston uses RMSE while Merton uses MSE
- Heston uses MSE while Merton uses RMSE
- They use the same metric

Question 6

In the brute force optimization ranges of Lesson 2 (Merton model), what range is specified for the jump intensity λ?

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- (0.10, 0.301, 0.1)
- (-0.5, 0.01, 0.1)
- (0.10, 0.401, 0.1)
- (0.075, 0.201, 0.025)

Question 7

In the Lewis integral for Merton, why is the upper bound set to 50 instead of infinity?

- Lewis method requires bounds under 100
- It matches the number of options
- Merton requires finite bounds
- Practical truncation as the integral converges quickly

Question 8

How does the integral evaluation differ between Merton and more complex models?

- There's no difference

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- Complex models use Monte Carlo instead
- Merton doesn't need numerical integration
- Merton uses quad with limit 50, simpler than unbounded integrals

Question 9

What is the distribution of $\log(1 + J_t)$ in the Bates model?

- Exponential distribution
- Normal distribution $N(\log(1 + \mu_J) - \delta^2/2
- Poisson distribution
- Uniform distribution

Question 10

What does the SimpsonW variable implement in the FFT code?

- Correlation adjustment

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- Volatility scaling
- Simpson's rule weights for numerical integration
- Jump size weights

Question 11

What is the correlation between the trend component and jump component in Bates?

- ρ (rho)
- -1
- 0
- 1

Question 12

In the M76J_char_func, what is different from the original Merton characteristic function?

- Nothing is different
- Jump parameters are squared

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- The volatility term σ is removed
- The risk-free rate is doubled

Question 13

In the Tikhonov penalty term, what does $(p_0 - p)^2$ represent?

- Squared deviation from initial parameter values
- Volatility of volatility
- Error between model and market prices
- Jump intensity variance

Question 14

What is the sequential calibration procedure for Bates (1996)?

- Calibrate jumps first, then volatility, then full model
- Calibrate all parameters simultaneously

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- Calibrate stochastic volatility first, then jumps, then full model
- Calibrate full model then refine components

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QUESTIONS

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Question 15

What does 'local_opt' flag control in the error function?

- Whether to apply the Tikhonov penalty
- Integration limits
- Printing of results
- Use of local vs global optimization

Question 16

What are the two basic flaws in calibrating jump diffusion models?

- Slow convergence and high cost

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TOP

- Lack of data and model complexity
- Numerical instability and overflow
- Different parameter combinations yielding similar errors and many local minima

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